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Biobehavioral Influences of Stress and Inflammation on Mucositis in Adolescents and Young Adults with Cancer: Results from a Pilot Study

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Purpose: Chemotherapy-induced mucositis is a prevalent and burdensome toxicity among adolescent and young adults (AYAs) with cancer and impedes the delivery of optimal therapy. Its development is not well understood, but baseline stress and inflammation may be contributory factors. This pilot study evaluates stress and inflammation as risk factors for mucositis, identifies effect size estimates, and evaluates the feasibility of a prospective study to investigate mucositis development.

Methods: Thirty AYAs receiving chemotherapy with substantial risk of mucositis completed baseline stress measures, and serum was collected for inflammatory biomarker analysis. Regression and mediation analyses determined the relationship between stress/inflammation and mucositis.

Results: Stress appears to be a significant risk factor for incidence of mucositis (odds ratio 1.13, $p=0.125$) and predicts total mucositis score ($\beta=0.281$, $p=0.023$) as well as peak incidence ($\beta=0.052$, $p=0.018$). Baseline levels of interleukin (IL)-1 α and epidermal growth factor (EGF) predicted mucositis development, and EGF and IL-8 may mediate the relationship between stress and mucositis. Findings suggest that stress-induced inflammation exacerbates symptom development.

Conclusion: Results from this pilot study inform mucositis symptom models, suggesting that psychosocial and physiologic factors are involved in development. Importantly, this pilot study provides initial effect size estimates, including magnitude and direction of relationships, that are essential to informing larger, more robustly powered studies. High enrollment, low attrition, and minimal missing data in this study suggest this model is feasible for research in this population. Importantly, this work is a first step in identifying new risk factors for mucositis and targets for nurse-led interventions to prevent toxicity development.

Keywords: mucositis, stomatitis, adolescent and young adult oncology, pediatric oncology, chemotherapy toxicities

Introduction

CHEMOTHERAPY-INDUCED MUCOSITIS is a frequent adverse effect and burdensome toxicity of cancer-directed therapy.^{1–4} Patients cite mucositis as one of the most distressing effects of treatment because of its painful

presentation and impact on the ability to eat, drink, and speak normally.^{5,6} Subsequent malnutrition and dehydration further worsen quality of life and lengthen post-chemotherapy recovery,^{4,6,7} and mucositis develops concurrently with immunosuppression, creating a significant risk of infections.^{6,8,9}

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Development of severe mucositis necessitates de-escalation of treatment intensity, so a solitary incidence may change the course of treatment and compromise overall survival. Symptomatic care remains the only available management once the condition develops^{10–12} and a number of preventative approaches have been identified but these have not proven to be universally effective.^{13,14}

Chemotherapy alone does not induce mucositis¹⁵ but is the first step in ulcer formation. Chemotherapy inhibits cellular growth and differentiation, ultimately leading to cell death that initiates an inflammatory cascade that worsens damage and exacerbates lesions.^{16–19} Increased reactive oxygen species and damage-associated pattern molecules also recruit a swift inflammatory response^{16–19} that damages the endothelium, dissolves connective tissue, and impairs oxygenation and repair mechanisms.^{16–19}

Overall, chemotherapy is the instigating factor in mucositis development, but it is the patient's inflammatory response that contributes to development.¹⁵ Higher baseline inflammation has been suggested as a risk factor for mucositis during radiation.²⁰ Modulating the inflammatory response, therefore, may be an opportune target for preventative interventions.

Incidence is highest in the adolescent and young adult (AYA) population, and up to 43% of patients aged 15–39 develop dose-limiting mucositis.^{21–25} The higher incidence of severe mucositis is particularly concerning since AYAs have seen the slowest improvements in cancer survival over the past 20 years.²⁶ Focused attention on prevention of dose-limiting toxicities in this cohort of oncology patients is particularly important. Unique to other age groups, AYAs also report the highest rates and severity of stress during treatment,^{27–31} which is a known cause of inflammation.^{32,33}

Since inflammation has a known role in mucositis, stress may also be contributing by way of stress-induced inflammation, priming patients for mucositis development by elevating inflammatory markers that exacerbate ulcer development. Animal models suggest that acute stress-induced inflammation significantly increases the risk of mucositis.^{34,35} It is not understood whether this relationship also exists in humans, and clinical studies of this sort have yet to be conducted. The purpose of this pilot study is to conduct initial exploration of associations between stress, inflammation, and mucositis among AYAs receiving chemotherapy and evaluate whether inflammation mediates the relationship between stress and mucositis.

Methods

This prospective longitudinal study methodology is available in detail in a separate publication,³⁶ but in brief, recruited AYAs receiving chemotherapy from two hospitals in the mid-Atlantic region of the United States. The Institutional Review Boards (IRBs) at each center provided approval for this study (Johns Hopkins IRB No. 00245176). To be eligible, participants must have been between the ages of 15 and 39 years and receiving chemotherapy with $\geq 20\%$ incidence of dose-limiting mucositis.

Participants were excluded if they had a documented cognitive deficit, used anti-inflammatory medications, or had an infection requiring antibiotics in the past 14 days as these

would interfere with the measure of stress and inflammation respectively. Participants were enrolled for data collection during a single cycle of eligible chemotherapy.

Recruitment and data collection

The study team approached eligible participants for enrollment, advertised the study through the electronic medical records system, and accepted referrals from clinicians. Participant responses were collected through Research Data Capture surveys emailed to the participant; demographic and clinical data were extracted from the medical record.

Measures

Demographic and clinical data. Age, sex, chemotherapy agent(s) and doses, diagnosis, and absolute lymphocyte count at the time of chemotherapy were extracted from the medical records. Participants self-reported their race/ethnicity. Lymphocyte count was collected to account for immunosuppression, which has a known influence on mucositis development. The risk of mucositis for each participant was determined by reviewing toxicity profiles from past clinical trials^{37–39} and used to dichotomize participants as “high” (20%–25% chance) or “very high” (>25%) risk of developing dose-limiting mucositis.

Stress and self-efficacy. The National Institutes of Health (NIH) Perceived Stress Scale was used to measure baseline stress at the beginning of chemotherapy visits. Because self-efficacy has been identified as a potential moderating factor for symptoms in AYAs with cancer,⁴⁰ the NIH Self-Efficacy measure was also administered. Validated and reliable versions of each measure are available for ages 13–17 and 18+, have high internal consistency ($\alpha > 0.89$),^{41,42} and comparative fit index exceeding 0.89.⁴¹ Large-sample normative means allow for the calculation of *t*-scores, which accounts for variance by age.^{41,43,44} Stress scores were analyzed as continuous predictors and to identify participants who were categorized as “high stress” (>1 standard deviation [SD] above normative mean).

Inflammatory markers. Inflammatory markers were measured via serum collected before chemotherapy administration drawn with clinical labs. Biomarkers hypothesized in the relationship between stress and mucositis included epidermal growth factor (EGF), tumor necrosis factor alpha (TNF α), interferon gamma (IFN γ), interleukin (IL)-1(a and b), IL-6, and IL-12.^{34,35} Additional exploratory markers included granulocyte-monocyte stimulating factor (GM-CSF), monocyte chemoattractant proteins 1 and 3 (MCP1, MCP3), vascular endothelial growth factor (VEGF), and IL-8, IL-10, IL-12p40, IL-12p70, IL-13, and IL-18.

Biomarkers were analyzed using the Milliplex MAP Human High Sensitivity multiplex assay quantifying levels of each in pg/mL. Each sample was tested in duplicate, and the mean value was used for analysis. The value of half the lower limit of detection was used for samples that had inflammatory marker levels below the assay's lower detection level.⁴⁵

Mucositis symptoms. Participants reported symptoms daily over the 2-week period after chemotherapy via the Patient-Reported Outcomes version of the Common Terminology

Criteria for Adverse Events (PRO-CTCAE) mucositis grading scale.⁴⁶ The PRO-CTCAE has high validity and reliability over a recall period of up to 7 days for AYAs and correlates to the traditional clinician-assigned CTCAE severity through a single-item 0–4 scale.^{46–48} For analysis, mucositis was evaluated in three ways: (1) incidence, dichotomizing participants who reported no symptoms versus those who reported any; (2) peak severity determined by highest intensity for each participant; and (3) total sum score across all 14 days accounting for severity and duration.

Analysis

Frequencies, percentages, means, and SDs were used to describe demographic and clinical characteristics. Linear regression evaluated influences of age on stress and inflammation. Independent *t*-tests explored differences in inflammation between groups based on sex, race, diagnosis, high stress, and mucositis incidence. Chi square evaluated mucositis incidence between groups based on chemotherapy, diagnosis, and high stress. Regression models evaluated relationships between stress, inflammation, and mucositis. An alpha level of 0.05 guided interpretation; however, given the small sample size, greater emphasis is placed on effect size.

Structural equation modeling evaluated inflammatory markers as mediators between stress and mucositis (Fig. 1). Logistic regression was used for the bivariate outcome of mucositis incidence. Mediation was suggested if the mediation model *p* value was lower than that of the direct model. Sobel's test evaluated mediation for outcomes of peak severity and total score.

Results

Thirty participants were recruited over a 14-month period ending in November 2021. One did not complete the stress survey, and another could not have biomarkers analyzed due to sample hemolysis. Ages ranged from 15 to 34 with a mean of 19.3 years (SD 4.8 years), and this sample equivocally represents male and female participants (Table 1). Two thirds (66.7%) reported their race as White, and 33.3% identified as Black/African American. Diagnoses included sarcoma (36.7%), lymphoma (33.3%), leukemia (20%), and other tumors (10%) translating to a variety of chemotherapy regimens (Table 1).

Mean stress *t*-scores for this population were higher than normative scores ($p < 0.002$), and 11 participants' (37.9%) stress scores met criteria for high stress (>1 SD above normative mean). Stress did not vary by age ($\beta = 0.269$, $p > 0.1$) and was not different based on sex or race ($p > 0.8$).

Sixteen participants (53.3%) developed mucositis at some point during the study period, and the remaining 14 (46.7%) did not report any symptoms (Table 1). Eight (26.67%) had peak severity described as mild, and 5 (16.67%) developed moderate pain. Three participants (10%) developed dose-limiting toxicity: two with grade 3 (severe pain) and one with grade 4 (very severe pain). The mean total mucositis score was 3.83 (SD 5.87). Mucositis incidence, severity, and total score were not significantly different with regard to participant sex, age, or oncology diagnosis.

Relationships between stress, inflammation, and mucositis

Stress-induced inflammation. Black participants had higher GMCSF (74.24 vs. 6.1, $p = 0.036$) and VEGF (941.22 vs. 302.02, $p = 0.002$), and patients with leukemia had higher IL-8 (17.02 vs. 7.65, $p = 0.029$) and MCP1 (1000.73 vs. 526.76, $p = 0.011$). Therefore, race and diagnosis of leukemia were included as confounders in regression models predicting inflammation. In these analyses, stress only predicted levels of IL-8 ($\beta = -0.297$, $p = 0.040$), but not in the hypothesized direction (Table 2).

Inflammation-induced mucositis. Regression models and Chi square analyses did not suggest that mucositis incidence varied by age, race, or diagnosis ($p \geq 0.2$) but the odds of developing mucositis decreased with higher lymphocyte counts (odds ratio [OR] 0.99, $p = 0.026$). Incidence was higher for patients who received high-risk chemotherapy (71%) compared with lower risk chemotherapy (37.5%, Chi square $p = 0.06$), and total mucositis scores were higher for those receiving high-risk chemotherapy compared with those receiving low-risk chemotherapy (5.14 vs. 2.69, $p = 0.26$). Therefore, lymphocyte count and chemotherapy regimen were included as confounders in analyses predicting mucositis as well as self-efficacy due to findings from past research.⁴⁰

Table 3 displays results of regression models predicting mucositis from inflammatory markers while controlling for the confounders cited earlier. IL-1a levels predict mucositis

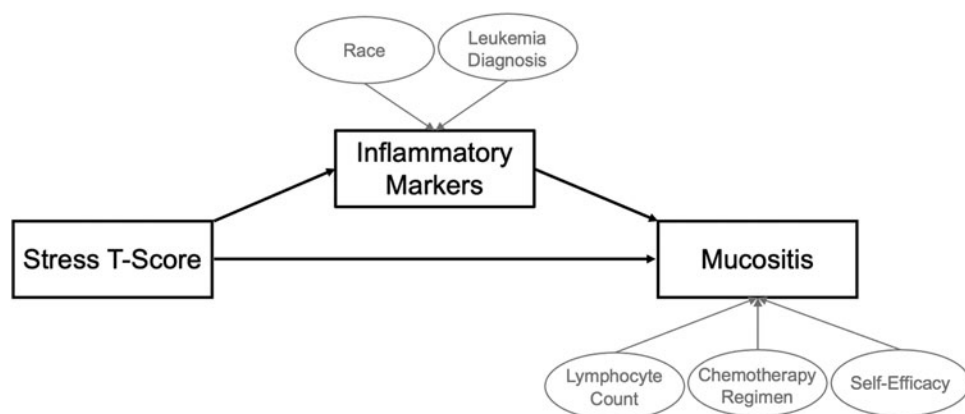


FIG. 1. Structural equation modeling. Black squares identify independent, mediating, and outcome variables; gray ovals identify confounding variables.

TABLE 1. DEMOGRAPHICS
AND CLINICAL CHARACTERISTICS

Age (mean years)	19.3 (SD 4.8)
Sex, <i>n</i> (%)	
Male	14 (46.7)
Female	16 (53.3)
Race, <i>n</i> (%)	
White	20 (66.7)
Black	10 (33.3)
Diagnosis, <i>n</i> (%)	
Leukemia	6 (20)
Sarcoma	11 (36.7)
Lymphoma	10 (33.3)
Other	3 (10)
Chemotherapy, <i>n</i> (%)	
Ifosfamide and etoposide	7 (23.3)
Doxorubicin	4 (13.3)
Methotrexate	4 (13.3)
Cyclophosphamide and doxorubicin	4 (13.3)
Cyclophosphamide, doxorubicin, and etoposide	3 (10)
Cytarabine	2 (6.7)
Cytarabine and daunorubicin	2 (6.7)
Cyclophosphamide and fludarabine	2 (6.7)
Cytarabine and etoposide	1 (3.3)
Cyclophosphamide and etoposide	1 (3.3)
Stress (mean <i>t</i> -score)	55.6 (SD 11.4)
Stress categories, <i>n</i> (%)	
High stress (score ≥ 1 SD above normal)	11 (37.9)
Normal stress (score ± 1 SD within normal)	19 (63.3)
Mucositis risk, <i>n</i> (%)	
High (20–25% chance)	16 (53.3)
Very high (>25% chance)	14 (46.7)
Mucositis incidence, <i>n</i> (%)	
No mucositis symptoms	16 (53.3)
Mucositis symptoms present	14 (46.7)
Peak mucositis severity, <i>n</i> (%)	
Grade 0: No symptoms or pain	14 (46.67)
Grade 1: Mild pain, no need for pain medication	8 (26.67)
Grade 2: Moderate pain, but no change in oral intake	5 (16.67)
Grade 3: Severe pain, stopped eating or drinking normally	2 (6.67)
Grade 4: Very severe pain, had to go to hospital or clinic	1 (3.33)
Total mucositis score (mean)	3.83 (SD 5.87)

SD, standard deviation.

incidence (OR 2.66, $p=0.084$); EGF levels predict total mucositis score ($\beta=-0.024$, $p=0.030$) and peak severity ($\beta=-0.004$, $p=0.025$). The remaining inflammatory markers did not have significant relationships with mucositis development.

Stress-induced mucositis. As with models predicting mucositis from inflammation, lymphocyte count, self-efficacy, and chemotherapy regimen were included in models predicting mucositis from stress (Table 3). Stress was not a reliable

 TABLE 2. PREDICTING INFLAMMATION FROM STRESS,
CONTROLLING FOR RACE AND LEUKEMIA DIAGNOSIS

Biomarker	β	p
EGF	−0.566	0.741
GMCSF	−1.502	0.246
IFN γ	−0.786	0.268
IL-1a	−0.102	0.116
IL-1b	−0.039	0.199
IL-6	−0.061	0.335
IL-8	−0.297	0.040
IL-10	−0.054	0.667
IL-12p40	−0.259	0.407
IL-12p70	−1.00	0.238
IL-13	−1.73	0.328
IL-18	−2.77	0.390
MCP1	−1.47	0.826
MCP3	−0.141	0.839
TNF α	−0.562	0.448
VEGF	−2.98	0.701

β , standardized beta-coefficient; EGF, epidermal growth factor; GMCSF, granulocyte-monocyte stimulating factor; IFN γ , interferon gamma; IL, interleukin; MCP, monocyte chemoattractant protein; TNF α , tumor necrosis factor alpha; VEGF, vascular endothelial growth factor.

predictor of mucositis incidence (OR 1.13, $p=0.125$) but it did predict total score ($\beta=0.281$, $p=0.023$) and peak mucositis severity ($\beta=0.052$, $p=0.018$) with higher baseline stress, again correlating with higher severity and a longer duration.

Mediation analysis

Structural equation modeling (Fig. 1) evaluated the indirect effect that each inflammatory marker had on mucositis development individually (Table 4). Several markers

 TABLE 3. PREDICTION OF MUCOSITIS
FROM INFLAMMATORY MARKERS AND STRESS,
CONTROLLING FOR LYMPHOCYTE COUNT,
SELF-EFFICACY, AND CHEMOTHERAPY

Biomarker	Mucositis incidence		Total mucositis		Peak mucositis	
	OR	p	β	p	β	p
EGF	0.997	0.583	−0.024	0.030	−0.004	0.025
GMCSF	0.994	0.467	−0.008	0.555	−0.002	0.523
IFN γ	0.992	0.638	−0.012	0.656	−0.002	0.619
IL-1a	2.66	0.084	0.229	0.509	0.056	0.362
IL-1b	1.00	0.996	−0.689	0.307	−0.092	0.442
IL-6	1.02	0.950	0.134	0.693	0.021	0.727
IL-8	1.02	0.715	−0.134	0.338	−0.018	0.463
IL-10	1.03	0.648	−0.043	0.782	−0.014	0.614
IL-12p40	0.957	0.192	0.063	0.301	0.005	0.657
IL-12p70	0.993	0.596	−0.017	0.451	−0.003	0.465
IL-13	0.996	0.545	−0.006	0.583	−0.001	0.563
IL-18	1.00	0.857	0.004	0.465	−0.001	0.526
MCP1	0.997	0.285	0.001	0.596	−0.001	0.662
MCP3	0.983	0.230	−0.013	0.636	−0.003	0.477
TNF α	0.990	0.484	−0.024	0.325	−0.004	0.339
VEGF	1.00	0.722	−0.002	0.451	−0.001	0.626
Stress	1.13	0.125	0.281	0.023	0.052	0.018

Bold values indicate $p<0.1$.

OR, odds ratio.

TABLE 4. INDIRECT EFFECTS OF INFLAMMATION ON RELATIONSHIP BETWEEN STRESS AND MUCOSITIS

Biomarker	<i>Mucositis incidence</i>		<i>Total mucositis</i>			<i>Peak mucositis</i>		
	β	p	Coefficient	p	% effect	Coefficient	p	% effect
EGF	0.124	0.137	0.009	0.763	3.7	0.002	0.763	3.2
GMCSF	0.185	0.089*	0.000	0.922	—	0.000	0.910	—
IFN γ	0.183	0.091*	0.000	0.953	—	0.000	0.950	—
IL-1a	16.1	1.00	0.001	0.903	—	0.000	0.889	—
IL-1b	0.190	0.094*	0.001	0.891	—	0.000	0.858	—
IL-6	0.184	0.094*	0.000	0.936	—	0.000	0.932	—
IL-8	0.162	0.133	0.023	0.488	8.1	0.002	0.779	2.9
IL-10	0.194	0.108	0.000	0.955	—	0.000	0.952	—
IL-12p40	0.182	0.092*	0.001	0.901	—	0.000	0.895	—
IL-12p70	0.187	0.086*	0.000	0.975	—	0.000	0.975	—
IL-13	0.190	0.083*	0.000	0.996	—	0.000	0.996	—
IL-18	0.181	0.106	0.000	0.962	—	0.000	0.798	—
MCP1	0.189	0.108	0.000	0.974	—	0.000	0.974	—
MCP3	0.182	0.083*	0.000	0.918	—	0.000	0.885	—
TNF α	0.187	0.084*	0.000	0.925	—	0.000	0.906	—
VEGF	0.158	0.101	0.004	0.698	1.4	0.000	0.801	—

% effect indicates the percentage of effect of stress that is mediated by inflammatory marker, only displayed if >1%.

*Indicates improvement in prediction with inflammatory markers (p value reduces with addition of mediators).

(GMCSF, IFN γ , IL-1b, IL-6, IL-12p40, IL-12p70, IL-13, MCP3, and TNF α) improved mucositis incidence prediction to some degree. Sobel's test of mediation identified that IL-8 explained 8.1% of the effect of stress on total mucositis score as did EGF (3.7%) and VEGF (1.4%). Similarly, EGF and IL-8 mediated the relationship between stress and peak mucositis severity (3.2% and 2.9% of relationships, respectively).

Discussion

This study aimed at describing the initial relationships between stress, inflammation, and mucositis in AYAs and at evaluating whether inflammation mediates this relationship. Findings provide initial effect sizes and identify valuable new insights into the complex processes that contribute to mucositis.

Stress and inflammation as novel predictors of mucositis

Our results echo those of animal models and suggest that stress may serve as a risk factor for chemotherapy-induced mucositis. The odds of developing mucositis increased by 13% for every 1-unit increase in baseline stress score ($p=0.125$) in this study; although it was not powered to identify conclusive relationships, it demonstrates that the relationship is in the theorized direction. Stress does become a significant predictor for mucositis severity and duration; peak severity was reliably predicted by baseline stress ($\beta=0.052$, $p=0.018$) as well as total score ($\beta=0.281$, $p=0.023$).

Stress is a component of the complex psychosocial profile of young people with cancer and a modifiable factor in this age group.⁴⁹ Effective interventions that improve stress in AYAs have been identified^{49–51} but are not universally implemented. This pilot study suggests that stress may be an important modifiable risk factor for mucositis. Reductions may mitigate mucositis and ease the burden of cancer care. Larger, more robustly powered studies will be needed to

better elucidate the role of stress in mucositis more clearly by more adequately controlling for mucositis risk due to chemotherapy agent(s) and dose.

This study was also not able to adequately evaluate whether inflammation mediated the relationship between stress and mucositis due to sample size limitations but identified several markers that may function in this role, including EGF, GMCSF, IFN γ , IL-1b, IL-6, IL-8, IL-12p40, IL-12p70, IL-13, MCP3, TNF α , and VEGF, which should also be evaluated in future work.

Inflammatory biomarkers as mediators of stress-induced inflammation. In addition, this pilot study has identified that IL-1a and EGF are two inflammatory markers that may serve as risk factors for mucositis development.

IL-1a is a potent stimulator of acute and chronic inflammatory processes^{52–54} and an important component of cell homeostasis in mucosal tissues.⁵⁴ Production increases in response to growth factors, other pro-inflammatory cytokines, and stress.^{52,54} In this pilot study, the hypothesis that IL-1a levels increased in response to stress in AYAs with cancer was not supported. However, the hypothesis that higher baseline levels correlate with mucositis development hold true.

Higher levels of IL-1a at the time of chemotherapy administration were associated with a 2.7-fold increase in mucositis incidence when controlling for relevant clinical factors ($p=0.084$). Considering our small sample size, it appears as though IL-1a levels could emerge as a predictor of mucositis and possible target for prevention. However, IL-1a did not reliably predict total mucositis score or peak mucositis score, suggesting that this relationship with mucositis remains incompletely understood and requires additional research to more comprehensively understand.

The EGF is a growth factor that is involved in the homeostasis of mucosal tissues leading to increased growth and differentiation of epidermal and mucosal cells.^{55–58} Expression and upregulation leads to more durable cells, improved

intestinal barrier integrity, and higher cell proliferation.⁵⁸ The role of stress-induced changes to EGF concentration is not well understood,⁵⁹ but it is known that EGF attenuates the inflammatory response through anti-inflammatory actions.^{60–62}

In this pilot study, EGF levels reliably predicted total mucositis score ($\beta = -0.024$, $p = 0.030$) and peak severity ($\beta = -0.004$, $p = 0.025$). Although these effect sizes are small, findings support the hypothesis that inflammatory profiles contribute to mucositis and higher levels of this anti-inflammatory marker were protective against mucositis development. Before applications in the clinical setting, this would also require additional investigation to better elucidate the magnitude of these relationships.

Relevance to clinical practice and research. Initial findings suggest that AYAs with cancer may benefit from early interventions to reduce baseline stress and subsequent stress-induced mucositis. Clinical services in this domain are scarce; in a national survey of 142 pediatric oncology palliative care clinicians, only one-third (33.3%) of these programs had a psychologist, 19% had an art therapist, and only 7.5% had a psychiatrist. Further, only 35% had an AYA program available for patients.⁶³

These services, which are intended to assist with symptom management, remain composed primarily of medical staff and are not well prepared to address the psychosocial needs of AYAs. This pitfall in health care delivery may partially explain the higher stress rates for AYAs. Improved clinical care delivery that addresses the psychosocial needs of this population might serve to reduce symptom profiles by way of improved stress profiles. By demonstrating the broad-reaching impacts of improved psychosocial care, clinicians may be in a better position to advocate for these interdisciplinary services.

Modulation of the inflammatory profile through the use of anti-inflammatory agents may also improve mucositis development,^{64–67} but further clinical studies need to be conducted to more definitively determine their roles in this regard. These include the use of anti-inflammatory mouthwashes, which impact local inflammatory profiles that are limited to the oral cavity. Since this study evaluated systemic inflammatory markers via whole blood, studies evaluating salivary inflammatory markers may also assist with determining the impacts of localized inflammatory profiles on mucositis development.

Limitations and strengths. This prospective pilot study has several limitations due to the design and included variables. The small sample size introduces risk for type I and type II statistical errors while performing multiple analyses with similar outcomes. Attempts were made to consistently collect inflammatory markers immediately before chemotherapy, but lab draws for this study were performed with clinical draws; thus, there are variations in the timing of collection, and all were drawn before chemotherapy administration.

The use of half the lower limit of detection for samples that had undetectable levels did not accurately represent the true variability in the inflammatory markers. It should be noted that this study was conducted during the COVID-19 pandemic, and discussions of school/university closures were in national discourse. This social environment likely influenced

baseline stress levels in AYA participants since it complicated their typical social environment and clinical care. Lastly, since this was an observational study, there was no standardization of oral care for participants, which may have impacted mucositis development.

Irrespective of these limitations, the pilot study has strengths in its prospective design, allowing for the evaluation of causal relationships to be inferred from these findings. It is the first to evaluate these relationships in humans and provides effect size estimates that will inform larger studies that can better address the limitations of this work. The inclusion of participant-reported symptoms provided a consistent measure for each participant and is consistent with recommendations for symptom assessment in this population.^{5,68} Only one participant approached for this study declined enrollment, citing that it seemed too burdensome during therapy; however, the high enrollment rate and few missing data suggest that this model is appropriate for this population.

Recommendations for future research. This study controlled for the effect of chemotherapy on mucositis by stratifying participants into risk categories, but it may be more appropriate to stratify participants based on chemotherapy agent and control for the dose administered to better manage these confounders. A larger sample will better elucidate the inflammatory profile of these patients while considering variations due to diagnosis, demographics, and age-related changes and provide statistical power to detect true relationships in this regard. Exploration of salivary inflammatory markers may also hold promise as a means to better understand mucositis development in this population.

Conclusions

This pilot study sought to estimate effect sizes for the relationships between stress, inflammation, and mucositis and to investigate stress and inflammation as risk factors for mucositis development. Findings suggest that stress, IL-1a, and EGF at the time of chemotherapy administration may reliably predict mucositis development, but the relationships between stress and inflammation remain incompletely understood. Stress and inflammation might be important factors to address as a means to reduce mucositis development.

Results provide important insights for the future direction and design of clinical studies to better examine biobehavioral predictors of mucositis development and identify new targets for interventions to prevent mucositis.

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